

DATA 266 : LAB 1

Part 1: Building Large Language Model (LLM) from Scratch (30 Marks)

The objective of this question is to enable you to **design, implement, and train a GPT-style autoregressive language model from scratch**, based on the Transformer architecture from **Attention Is All You Need** paper <https://arxiv.org/pdf/1706.03762>
You will also analyse **common failure modes in sequence generation** to understand model behaviour and limitations.

Dataset:

- You must use a **subset of the TinyStories dataset**, designed for training small-scale language models.
- Link to research paper of the dataset: <https://arxiv.org/pdf/2305.07759>
- Link to dataset: <https://huggingface.co/datasets/roneneldan/TinyStories>

Actionable Tasks:

1.1. Data Preprocessing (5 Marks)

1. Load the dataset and tokenize the text at either the **character level**.
2. Create fixed-length input–target sequences suitable for autoregressive language modelling.
3. Convert characters/words into integer encodings (e.g. using `char_to_idx` and `idx_to_char` dictionaries)
4. Split the dataset into training (train: 100K) and validation (val: 10K) sets.

1.2. GPT Model Implementation (15 Marks)

1. Implement one or more **Transformer blocks** from scratch, each consisting of:
 - Multi-Head Self-Attention
 - Layer Normalization
 - Feed-Forward Network
 - Residual connections
2. Use **causal masking** in self-attention to ensure the model does not access future tokens during training.
3. Create **learnable embedding layers** for:
 - Token embeddings
 - Positional embeddings

4. Add a **language modelling head** that projects the final hidden representations to the vocabulary size for next-token prediction.

1.3. Training and Text Generation (8 Marks)

1. Train the model using **cross-entropy loss**.
2. Implement learning rate warm-up and scheduling.
3. Train the model for a minimum of 10 epochs.
4. Plot and report training and validation loss curves.
5. Generate text samples using greedy decoding or temperature-based sampling.

1.4. Sequence Model Failure Analysis (2 Marks)

You will analyse **three failure cases** observed in the generated text. For each failure case, you must:

- Provide the generated text snippet you observe as a failure case.
- Identify the type of failure (e.g., repetition, broken grammar, loss of coherence, hallucination) along with your observation.

*Use of prebuilt Transformer or attention modules is **not permitted**.*

Part 2: NLP Sentiment Analysis on IMDB Movie Reviews (30 Marks)

The objective of this question is to enable you to perform **binary sentiment classification** using supervised Natural Language Processing techniques, compare multiple models, and critically analyse their performance using standard metrics.

Dataset:

- You must use the **IMDB Movie Review dataset**
- Link to dataset: <https://huggingface.co/datasets/stanfordnlp/imdb>

Actionable Tasks:

2.1. Data Preprocessing (10 Marks)

1. Analyse review length distribution, class distribution and class balance.
2. Handle missing values or malformed entries, if any.
3. Perform text preprocessing, including:

- lowercasing
 - removal of punctuation and special characters
 - stopword removal
 - stemming or lemmatization (if applicable)
4. Tokenise the processed text.
 5. Create **textual embeddings learned from scratch**.

2.2. Model Training and Evaluation (15 Marks)

You will train **at least two models** for sentiment classification:

- **Model 1 (Baseline Model):** Implement a standard sentiment classification model to establish baseline performance.
- **Model 2 (Experimental Model):** Implement a second model that attempts to improve upon the baseline through architectural or hyperparameter changes.

For **each model**, you must:

1. Justify the model architecture and embedding choices.
2. Evaluate performance using:
 - Accuracy
 - Precision
 - Recall
 - F1-score
 - Confusion Matrix

2.3. Comparative Analysis and Observations (5 Marks)

- Compare both models using above metrics.
- Discuss observed strengths, weaknesses, and limitations.
- Briefly comment on possible improvements or future work.

*Use of pre-trained embeddings or pre-trained language models is **not permitted**.*

Part 3: Image Style Transfer using Generative Adversarial Network (GAN) - Kaggle Class Competition (30 + 10 Marks)

In this part, you will participate in a **Kaggle Class Competition** and implement a **Generative Adversarial Network (GAN)** for **image style transfer** between two visual domains.

Your objective is to train a model that learns to translate images from one domain to another **without paired examples**, while preserving content and visual realism.

Kaggle Class Competition Signup Link :

- Sign Up Link: <https://www.kaggle.com/t/61ced43d0ddc40b09887c7b0de77273f>
- Model predictions will be evaluated, and rankings will be displayed on the leaderboard:

Leaderboard Rank	Points Awarded
1 - 3	10
4 - 6	9
7 - 9	8
10 - 12	7
13 +	6

- Link to relevant paper: <https://arxiv.org/pdf/1703.10593>

Actionable Tasks:

3.1 Model Implementation and Training (20 Marks)

1. Prepare two **unpaired image domains** from the dataset.
2. Implement a **CycleGAN architecture**, consisting of:
 - Two generators
 - Two discriminators
3. Train the model using:
 - Adversarial loss
 - Cycle-consistency loss
4. Perform image translation between the two domains.
5. Analyse training behaviour, including convergence and stability.

3.2 Evaluation and Analysis (10 Marks)

1. Visual quality of translated images.
2. Correct implementation of cycle-consistency constraints.
3. Training stability and loss behaviour.
4. Quality of analysis, observations, and discussion of shortcomings

Brief submission guidelines for this part will be posted separately on Canvas.
